

ABDULRAHMAN ALSAADI

MECHATRONICS & ROBOTICS ENGINEER

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PROFESSIONAL SUMMARY

Mechatronics and robotics engineer focused on developing complete robotic systems from mechanical design and embedded control through software integration and physical testing. Hands-on experience building robotic manipulators, integrating actuators and controllers, and taking robotics projects from CAD and simulation to functional prototypes.

TECHNICAL SKILLS

Robotics & Software: ROS 2 Humble, Gazebo, RViz, URDF/Xacro, ros2_control, C++, Python, Linux (Ubuntu), Git

Autonomy & Computer Vision: OpenCV, MAVLink, pymavlink, ArduPilot SITL, MQTT

Embedded Systems & Hardware: Raspberry Pi 5, ESP32, Arduino, Cube Orange, UART, PWM, stepper/servo/BLDC motors

Mechanical Design & Analysis: SolidWorks, ANSYS, FEA, Topology Optimization, 3D Printing/DFM, Manufacturing Drawings

SELECTED ENGINEERING PROJECTS

WD DRONE - TEKNOFEST 2026, Rotary Wing Category

Karabük, Türkiye

Founding Team Member | Simulation & Systems Integration

Feb 2026 - Jul 2026

- Created a custom Gazebo/ArduPilot SITL environment to test autonomous waypoint survey, target approach, payload delivery, and RTL mission logic before physical flight testing.
- Integrated a Raspberry Pi 5, Raspberry Pi Camera Module 3, and Cube Orange flight controller through MAVLink/pymavlink; developed OpenCV-based color and geometric target detection at **~30 FPS** and generated velocity corrections from target center offsets.
- Demonstrated autonomous waypoint flight and AUTO/GUIDED/RTL transitions in flight; physically tested target detection and payload release across **five flight-test sessions** at approximately 3-10 m altitude.
- Verified propulsion and structural performance through a **1,500 g motor-thrust test**, an **8 kg static-load test**, a **12-minute unloaded hover**, and a **7-minute hover with a 1 kg payload**; coordinated a six-member team across mechanical, avionics, software, and computer vision.

SCARA ROBOTIC ARM - GRADUATION PROJECT

Karabük, Türkiye

Mechatronics & Robotics Engineer

Apr 2026 - Jun 2026

- Independently designed, 3D printed, and assembled a 3-DOF SCARA arm with a **492.5 mm reach**, vertical Z-axis, and servo gripper, applying DFM principles to print orientation, support reduction, tolerances, and bearing fits.
- Calculated joint torque to guide motor selection; configured NEMA 17 steppers with DRV8825 drivers, limit switches, bearings, and a belt/gear transmission; physically tested a **100 g payload at full extension**.
- Programmed the ESP32 in C++ for forward and inverse kinematics, homing, motion limits, stepper control, and gripper operation, then confirmed commanded motion through physical testing.
- Applied FEA and topology optimization, manufactured revised PLA+ links, and integrated the SCARA, mobile robot, and conveyor through a **three-node MQTT network** to demonstrate autonomous box transfer and shelving.

ROS 2 MYCOBOT 280 SIMULATION

Personal Project

ROS 2 Humble | Ubuntu 22.04

GitHub Project

- Configured and simulated a 6-DOF myCobot 280 with **ROS 2 Humble** using URDF/Xacro, RViz, Gazebo, and ros2_control.
- Implemented ROS 2 communication and control workflows using topics, publishers/subscribers, parameters, actions, and launch files; commanded manipulator motion through FollowJointTrajectory with joint_state_publisher and robot_state_publisher.

EXPERIENCE

SMART METHODS

Jeddah, Saudi Arabia

Full Stack Robotics Engineering Intern

Jul 2025 - Aug 2025

- Independently delivered **13 engineer-evaluated robotics projects** across mechanical design, embedded control, computer vision, AI, and robot interfaces; published the project repositories on GitHub.
- Modeled and analyzed a 5-DOF robotic arm, performed forward kinematics and joint-torque calculations, programmed Arduino/ESP32 pick-and-place applications, and built Python/OpenCV and database-backed robot-control interfaces.

EDUCATION

Bachelor of Science (B.Sc.) in Mechatronics Engineering

Jun 2021 - Jun 2026

Karabük University, Türkiye

CERTIFICATIONS & PROFESSIONAL MEMBERSHIPS

- Saudi Council of Engineers (SCE) - Engineer Membership (Active)
- Certified SOLIDWORKS Design Professional - Mechanical Design (CSWP), Dassault Systèmes | Aug 2026
- Certified SOLIDWORKS Associate — Mechanical Design (CSWA), Dassault Systèmes | May 2026
- ROS 2 Fundamentals, Karabük University | May 2025

LANGUAGES

Arabic - Native | English - Upper-intermediate | Turkish - Intermediate